

When Grasp Metrics Disagree: Contact, Load, and Retention in Simulation

Supplementary Material

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This supplement preserves the detailed V4 protocol, results and data-role boundaries behind the compact main paper. V4.5 is an editorial release: no new simulation, training, reference labeling or hypothesis test was introduced. Repeated numbers here describe the same units as the main paper, not an additional evaluation. Frozen identifiers B0–B7 and H1/H4 are retained so that the paper maps directly to the evidence records.

Contents

S1 Evaluation Targets and Instrument	1
S2 Frozen Experimental Design	2
S3 Results	4
S4 Detailed Controls, References and Uncertainty	6
S5 Historical Evidence and Candidate Library	7
S5.1 Preserved V3 Comparisons	8
S6 Numerical Qualification and Recovery	10
S7 Offline Reproduction and Raw-Evidence Scope	10
S8 Disclosure	11

S1 Evaluation Targets and Instrument

Common task conditions. For physical-step index t , let Q_t denote the source-specific task conditions, and let G contain unchanged episode-level eligibility conditions. Allegro retains its existing native outcome, settling boundary, lift threshold, thumb and opposing-region contact requirements, and prohibition on assistance or post-reset target-state edits. Its 10^{-4} N threshold means a specified region has a nonzero normal contact above a numerical floor; it does *not* prove that this contact carries the object’s weight. Other sources have different, explicitly declared Q (Table S1). We do not pool these tasks into a robot capability ranking.

Literal no-contact. Let n_t^E count target–environment contact records, including inactive records created by contact margins. With $T = 15$ s and a continuous interval I , the recorded contract is

$$C_{\text{NC}} = G \wedge \exists I : |I| \geq T, \quad \forall t \in I, Q_t \wedge (n_t^E = 0). \quad (1)$$

This is a legitimate representation-level requirement. Calling a rejection a false negative is meaningful only relative to a *different declared target*.

Realized no-external-load. Each contact supplies a local force f_i , torque τ_i , frame R_i , point p_i , and direction sign s_i chosen so the world wrench acts on the target. For target center of mass c , we compute

$$F_i = s_i R_i^\top f_i, \quad M_i = s_i R_i^\top \tau_i + (p_i - c) \times F_i, \quad (2)$$

and normalize the sum of individual magnitudes:

$$r_F(t) = \frac{\sum_{i \in E_t} \|F_i\| + d_F(t)}{mg}, \quad r_M(t) = \frac{\sum_{i \in E_t} \|M_i\| + d_M(t)}{mgL}. \quad (3)$$

Here L is the target half-diagonal, using a declared enclosing compound box for the mesh source. The damping terms d_F, d_M meter known target free-joint damping; they are not assumed zero. The model’s gravity is retained but is not itself classified as external support. Net-wrench cancellation is deliberately not used to establish absence of individual loads.

The frozen lower/upper bounds are 10^{-4} and 10^{-2} for both ratios. Below the lower bounds, a step is definitely unloaded only when relevant paths are accounted for and contact/geometry ambiguity is resolved. At or above either upper bound, it is definitely loaded. Intermediate loads, unexplained passive forces, and unresolved active-zero-load constraints remain unknown. Geometry uses a 10^{-5} m penetration tolerance; the Fetch adapter uses conservative enclosing geometry rather than claiming exact mesh distance. Known free-joint damping is evaluated at both the solve state and implicit endpoint, taking the larger magnitude. Equality, spring, fluid, friction-loss, or other unsupported paths do not silently pass.

Let I^- be the longest continuous definitely-unloaded Q interval and I^+ the longest possibly-unloaded Q interval. The no-load contract returns

$$C_{\text{NL}} = \begin{cases} \text{PASS}, & G \text{ holds and } |I^-| \geq T, \\ \text{FAIL}, & G \text{ fails or } |I^+| < T, \\ \text{INDETERMINATE}, & \text{otherwise.} \end{cases} \quad (4)$$

Telemetry validity is a separate field. Missing steps or inconsistent clocks are invalid data, not a physical failure. Returned intervals and reasons allow the decision to be inspected without relying on a selected frame.

Intervention-conditioned dependence. At a predeclared time t_0 , a full integration-state snapshot initializes two branches. The sham B_s preserves the model; B_e removes only permitted target–environment contact pairs. Both replay the same controls and mocap tape for $H = 3$ s. The endpoint Y requires all-sample displacement ≤ 0.03 m, rotation ≤ 0.35 rad, and speed ≤ 0.2 m/s relative to the initial palm-relative pose (world frame for the pad apparatus). We retain all four (Y_s, Y_e) cells: both retain, dependent $(1, 0)$, removal rescue $(0, 1)$, and both fail. In particular,

$$D_{\text{CF}} = 1 \iff Y_s = 1 \wedge Y_e = 0 \quad (5)$$

is conditional on that state, horizon, removal, and open-loop continuation. It is neither the historical no-load label nor proof of 15-second autonomous capability. The sham must match the original target position and velocity trace bit-for-bit. Unreachable or invalid branches remain in the registry.

Nonintrusive observation and reference. The observer reads each physical contact solve without adding a forward solve to the live trajectory. State restoration includes the complete `mjSTATE_INTEGRATION` state, not only position and velocity. Independent code reconstructs world wrenches from raw local records, geometry from saved transforms or primitive vertices, and continuous intervals with a separate loop. This detects implementation inconsistency, but shares MuJoCo and the task specification: it is not independent physical truth or human annotation.

S2 Frozen Experimental Design

Dynamics use MuJoCo [3, 7]; the Allegro model derives from the pinned Menagerie assets [4]. Development and final evidence are separated by an input-only protocol lock. Before generating final outcomes we froze code, sources, roles, seeds, distributions, thresholds, branch timing, reference sampling, candidate order, statistical scripts and resource budgets. All 89 qualification tests passed. The 80 development controls had identical complete integration-state traces with the observer enabled and disabled; source-specific qualifications and paired shams were checked separately. One implementer operated the workflow; hashing and timestamps do not establish independent blinding.

Table S1: Final units and task boundaries. Counterfactual and reference samples are subsets of these units. Counts must not be added as independent scenes.

Source	Units	Nominal task	Fixed branch / reference subset
Free-object controls	240	Eight apparatus groups, 30 initializations each; 18 s pregrasp retention, 15 s continuous Q after 1 s	48 states at 8 s; all 240 references
Allegro library	120	Six primitive families, 20 scenes each; unchanged repair K6 pool, 720 candidate schedules	20 rank-1 states at schedule end minus 4 s; 30 randomly sampled rank-1 references
Fetch expert	100	Native goal/reset and 50-action task; separate 500-action retention extension with lift and two-finger contact	20 states at 16 s; 30 random references
Shadow / DGB	100	Four held-out object identities, 25 library records each; native signed-force task separate from added 18 s gravity retention	20 states 2 s after gravity-on; 30 random references

Controls and exclusions. The control target is a free six-DOF body held by two actuated slide pads. There is no target guide or hidden stabilizer. The eight registered groups are hand-only, inactive-margin records, necessary floor, redundant floor, opposed environment contacts, intermittent floor, explicit applied assistance, and boundary clearance. Geometry, mass, friction and initial perturbations vary within locked ranges; initial signatures are disjoint from calibration. Half the intermittent group has periodic support and half a single 0.1 s support burst. Eight registered inactive-margin cases are rerun with only margin/gap changed; representation invariance is asserted only if their full integration traces match. These pairs are not eight new independent scenes.

The historical retrospective examines every candidate in 85 V3 scenes that passed a native criterion but had no no-contact acceptance. Of 510 candidate slots, 177 qualifying traces receive full-wrench replay; the remaining 333 fail unchanged common conditions. Unmeasured wrench values are not imputed. This retrospective is development evidence, not the final evaluation.

Baselines and ablations. B0 is the original native gate for Allegro. On the added apparatus/external retention tasks it is an accumulated- Q proxy, with native external outcomes reported separately. B1 requires contiguous Q ; B2 is C_{NC} ; B3 adds positive geometric clearance. B4 uses the same full-wrench thresholds, continuous time, passive-load accounting and uncertainty handling as B5, but no geometric guard. It is the primary simple baseline, not a weakened scalar-force substitute. B4-normal uses normal load without contact torque. B5 is the integrated instrument of Section S1.

B6 is static friction-cone/actuator feasibility: an inscribed eight-ray cone and bounded actuator generalized-force balance are solved by linear programming. Torsional/rolling support, dynamics and joint-limit support are not modeled; unsupported or unbounded actuation is marked not applicable. B7 applies six signed world-axis, one-weight forces for 1 s each, with gravity retained and environmental contact removed. These assess mechanical capacity or perturbed future behavior, not historical C_{NL} truth, and are evaluated only on the fixed counterfactual subset. B7’s six additional simulated seconds are charged. Single-factor B5 ablations remove geometry, torque, per-contact aggregation, uncertainty, or continuous-time accounting. Five operating points are registered for controls; none is chosen from final outcomes.

External scope. Allegro and Fetch use MuJoCo 3.8.0; the pinned DGB runtime uses 3.3.2. These are not independent physics engines. The Fetch controller receives its unchanged native goal; table-level goals are not forcibly raised. Its target damping is retained. The DGB task begins with the upstream pregrasp/grasp/squeeze sequence, then adds gravity-on retention; it is not learned tabletop acquisition. The native DGB model is environment-free, making withdrawal an audited no-op. Native robomimic execution was unavailable on this Windows host; the objective of evaluating an independent learned acquisition program is therefore unsupported, not silently replaced by DGB. Natural reference entries are simple random samples with inclusion probability 30/120 for Allegro rank-1 and 30/100 for each other source. Diagnostic examples, if used, are separate.

Candidate decisions and independent future use. The genuine version pool contains only baseline and repair; all development scores promote the same version, so version-selection route U1 stops. U2 uses the unchanged repair pool: choose the first PASS in the fixed K prefix, otherwise abstain, for $K \in \{1, 3, 6\}$.

Every method observes and is charged the same complete K schedules. All choices and score hashes are sealed before future outcomes are generated. Y_{use} is a separate 15 s continuation from the final state, with target–environment permission removed, last commands/mocap held fixed, and the all-sample palm-relative bounds above. It never reads an auditor verdict. All 720 candidate futures are evaluated for finite-pool regret; the decision denominator is still 120 scenes.

Statistical estimands. H1 compares B5–B4 correct acceptances per assigned control, subject to an empirical accepted-risk ceiling of 0.05. H4 compares usable selections per assigned Allegro scene at $K = 6$; abstentions remain in the denominator. The minimum practical gain is 0.05. Two confirmatory comparisons use two-sided 97.5% Bonferroni-adjusted percentile intervals from 10,000 paired scene bootstraps, stratified by fixed mechanism/family, seed 2026100101. Descriptive intervals are 95%. DGB descriptive resampling clusters by its four object identities. We report FPR, FNR, accepted risk, unknown/invalid counts and conservative bounds; risk is undefined with no accepted labeled cases. The fixed budget does not guarantee power for five-point gains or certify low population risk. A zero-discordance bootstrap is not an equivalence test.

S3 Results

Controlled reference agreement and incremental value. The final controls contain 77 no-load reference positives and 163 negatives. B4 and B5 have no observed disagreement with those labels. Their H1 difference in correct acceptance per assigned case is 0.0 percentage points, with paired 97.5% interval $[0.0, 0.0]$. The empirical risk condition is satisfied, but the tied comparison does not establish equivalence beyond these apparatus distributions. Positive geometry alone (B3) also matches the nominal labels. No added-geometry accuracy benefit is demonstrated. Relative to the no-load target, no-contact has 0 false acceptances and 56 false rejections among 77 positives; its different literal semantics remain intact (Table S2).

Which distinctions survive targeted tests? In 8/8 registered representation pairs the complete integration trace is bit-identical, while no-contact changes in 8. B4 and B5 do not change. Accumulated no-load accepts 92 controls versus 77 for continuous time; the additional acceptances arise in the one-burst intermittent group. The other nominal ablations do not establish an incremental benefit on this controlled set. The registered branch subset gives 33 both-retain, 6 dependent and 9 unreachable states. Loaded-but-retaining redundant supports separate realized load from dependence under this intervention. No empty four-cell category is filled by adding favorable cases.

Natural tasks expose limits rather than a general accuracy gain. The historical no-contact exclusions yield 0/85 rescued scenes under no-load. The new external results are source-specific (Table S3). Fetch succeeds in 100/100 native episodes, while added retention has 0 definite passes, 52 failures and 48 indeterminate outcomes. Retained target damping torque lies in the registered gray band; dropping torque would accept 48 cases but would change the intended target. This is not evidence that the native expert failed. The Shadow library yields 70 no-load passes, 29 failures and 1 indeterminate record; 1 record has invalid telemetry following a reproducible simulator instability. Its native force-test successes are 66/100. Withdrawal cannot test environmental dependence in this environment-free model.

Execution recovery and reference limits. 5 of 720 Allegro candidate records fail the frozen 10^{-10} m geometry-consistency assertion. A documented fail-closed recovery records them as invalid/indeterminate without widening the assertion, replacing cases, changing physics or exposing future-use scores. Their native outcomes are retained separately. This bookkeeping recovery was introduced after the first execution failure and is not described as pre-test code. The 30-entry probability reference samples per natural source remain separate from control labels; unknown reference cases are not called verified failures or successes.

Candidate acceptance does not by itself prove utility. At K6, B5 accepts 13 and yields 10 usable selections among all 120 scenes, with 107 abstentions. B4 and B5 return the same candidate or abstention on 120/120 scenes; their H4 difference is 0.0 points, paired 97.5% interval $[0.0, 0.0]$. No-contact and B5 differ in 0/120 K6 choices. Native-gate selection provides 15 usable choices from 40 acceptances. These secondary contrasts concern the added intervention-conditioned use, not improvements to the native program. Unknown candidate futures make the finite-pool oracle a lower bound, so the reported regret is an observed-pool diagnostic rather than true optimal regret.

Table S2: Final no-load reference comparison on controlled apparatus. B0/B1/B2 have their own valid semantics; errors below are relative to the fixed C_{NL} reference, not accusations that their original definitions were implemented incorrectly. U is indeterminate; invalid is separately reported in the text.

Method	Accepted	FP / 163	FN / 77	U	Accepted risk
B0	201	124	0	0	61.7%
B1	182	105	0	0	57.7%
B2	21	0	56	0	0.0%
B3	77	0	0	0	0.0%
B4_normal	77	0	0	0	0.0%
B4	77	0	0	0	0.0%
B5	77	0	0	0	0.0%
A_accumulated	92	15	0	0	16.3%

Table S3: External source tasks remain separate. Allegro rows here use rank-1 candidates across 120 scenes, not 720 independent trials. Native external task outcomes are not interchangeable with the added retention contract.

Source	N	Native	C_{NC} P	B4 P/F/U	B5 P/F/U	Invalid
Allegro rank-1	120	12	2	2/118/0	2/118/0	0
Fetch	100	100	48	0/52/48	0/52/48	0
Shadow / DGB	100	66	70	70/29/1	70/29/1	1

Table S4: U2 on all 120 fresh Allegro scenes. Usable is independent future retention of the selected candidate; abstentions remain in the denominator. All methods pay the same complete nominal K -prefix simulation budget.

Method	K1 accept/use	K3 accept/use	K6 accept/use	K6 abstain	Regret
B0	12 / 2	34 / 11	40 / 15	80	0.283
B1	11 / 2	30 / 11	36 / 15	84	0.283
B2	2 / 2	8 / 6	13 / 10	107	0.325
B3	2 / 2	8 / 6	13 / 10	107	0.325
B4_normal	2 / 2	8 / 6	13 / 10	107	0.325
B4	2 / 2	8 / 6	13 / 10	107	0.325
B5	2 / 2	8 / 6	13 / 10	107	0.325

Costs and scope of mechanical baselines. B4/B5 use the same nominal trajectories and no extra physical steps. Median offline scoring calls on controls take 2.02 and 1.88 ms, excluding instrumentation, IO and simulation. These timings are hardware-dependent and do not establish a runtime advantage. B6 and B7 are reported on the fixed subset, with solver time and extra probe steps charged; neither is relabeled as historical no-load ground truth. Appendix tables give their applicability and outcomes. Native DGB execution and staging were performed separately, but their step costs were not metered, so the recorded audited-segment budget is not presented as an exhaustive benchmark cost.

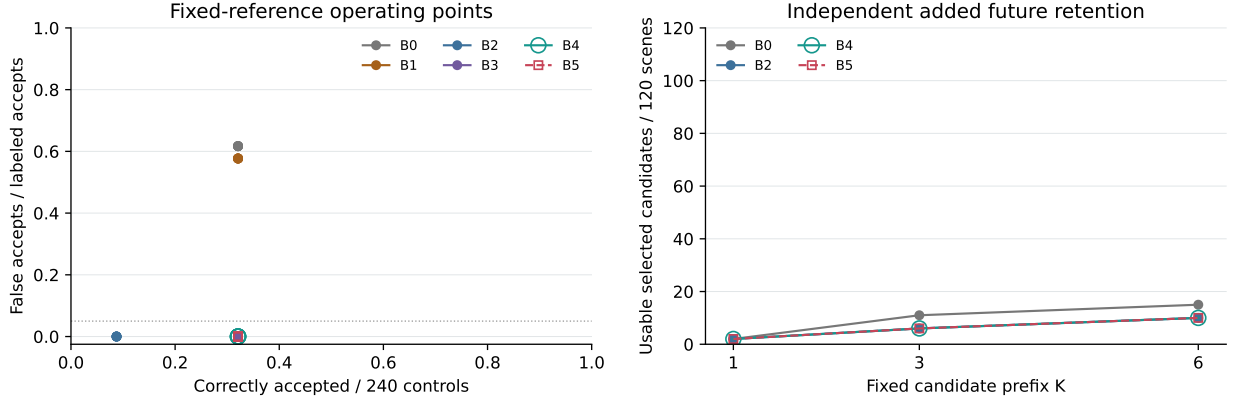


Figure S1: Registered control operating points and sealed candidate decisions. Curve points change both normalized force and torque bounds together relative to the fixed nominal reference. Overlapping B4/B5 curves must not be interpreted as independent performance gains. The utility panel reports all-scene counts.

S4 Detailed Controls, References and Uncertainty

Table S5: Final control pass counts by registered apparatus group.

Group	N	B0	B1	B2	B4	B5
applied assistance	30	0	0	0	0	0
boundary gap	30	30	30	0	26	26
inactive margin	30	30	30	0	30	30
independent	30	21	21	21	21	21
intermittent floor	30	30	11	0	0	0
necessary floor	30	30	30	0	0	0
opposing environment	30	30	30	0	0	0
redundant floor	30	30	30	0	0	0

Table S6: Fixed counterfactual subsets; unreachable/invalid states are not replaced.

Source	Both retain	Dependent	Rescue	Both fail	Unreached	Invalid
Controls	33	6	0	0	9	0
Allegro	0	1	0	0	19	0
Fetch	7	0	0	0	13	0
Shadow/dgb	16	0	0	0	4	0

Table S7: Mechanical baselines on the preselected branch subset. Valid and feasible/robust are separate counts; no-load truth is not their prediction target.

Source	Selected	B6 valid	B6 feasible	B7 valid	B7 robust
controls	48	39	39	39	33
allegro	20	0	0	1	0
fetch	20	0	0	7	7
dexgrasbench	20	0	0	16	14

The full confusion matrices, FPR/FNR, conditional error rates, accepted risk, unknown bounds and source-specific denominators are in the machine-readable analysis. For example, no acceptance means accepted risk is NA, not zero; a reference-indeterminate accepted item contributes uncertainty rather than a verified true positive. Binomial Wilson intervals are explicitly conditional-iid summaries, not corrections for shared apparatus or few object identities. The raw reconstruction is code-independent,

Table S8: Reference-sample labels and invalid telemetry. Allegro validity counts use 720 candidate executions, not independent scenes.

Source	Executions	Invalid	Ref N	Ref P	Ref F	Ref U
controls	240	0	240	77	163	0
allegro	720	5	30	2	28	0
fetch	100	0	30	0	15	15
dexgraspbench	100	1	30	21	9	0

not simulator-independent. Invalid Allegro candidate IDs from the geometry-consistency recovery are: 2ddb9ae96604e624158, 2658117c8f62d0ddf4c4, c1963347598f1be128f4, 555739848b81d855e167, a5eb27b988aecedb0b73.

S5 Historical Evidence and Candidate Library

The earlier reports [5, 6] provide development provenance, not new confirmation of this paper’s monitor comparisons. Their reused experimental history remains identifiable. The subsequent consolidated V3 experiment and present V4 study use different scene sets and analysis locks. The earlier in-house hand study’s 150/150 successes used an explicit assistance mechanism and are not strict no-stabilizer successes; the locked final strict evaluation was 0/150. Earlier open-hand validation reached 60/60 on the declared three-family suites, whereas later frozen blind suites yielded 38/120 and 50/120 for the successive program versions. Those historical splits and physical settings are not replaced or pooled with the present experiment.

In the preserved V3 compiled-primitive experiment, K6 native, contiguous and environment-clear counts were 128, 122, and 37 for baseline, versus 141, 133, and 48 for repair, each over 360 scenes. The 11 clear-support gains were spheres. Its actual uncertainty calculation was a paired scene bootstrap within six fixed family strata of 60 scenes, 5,000 draws, seed 2026093040, two-sided 95% percentile interval: 1.67–4.72 percentage points. Candidates and blocks were not separately resampled as independent scenes. All paired case keys were required to exist; no missing-record imputation was used. This historical analysis differs from the new registered 10,000-draw protocol.

On the same earlier 120 scenes, compiled primitive physics reduced no-contact success from 18 to 13 for baseline and 24 to 18 for repair. This changes geometry-dependent bounds and derived quantities as well as inertia, so it is not an isolated inertia intervention. Repair native totals remained 43 versus 43 while 7 cases gained and 7 lost success. Equal totals therefore did not mean case-level stability. A stricter definition is likewise not automatically more accurate for a different target.

The present development retrospective covers 85 excluded scenes and 510 candidate slots. Exactly 177 candidates were fully replayed with wrench telemetry; 333 already failed unchanged common conditions. The new contract rescues 0/85 scenes. The 333 unmeasured load histories are not claimed to have zero load or to have independent wrench-reference labels. The pool uses six candidates per scene; K=1 and K=3 are

Table S9: Candidate membership in the historical V3 360-scene pool. Shared/new/dropped refer to distinct complete candidate specifications by family, not new independent program checkpoints.

Family	Changed scenes	Shared	New	Dropped	Order / runner
bar	0/60	11	0	0	nearest XY / hold
capsule	0/60	12	0	0	nearest XY / hold
cube	0/60	6	0	0	fixed / staged
cylinder	0/60	12	0	0	nearest XY / hold
flat box	0/60	6	0	0	fixed / staged
sphere	46/60	12	6	0	nearest XY / hold

prefixes. There are no repeated candidates within a scene/policy prefix. No-extra equals baseline in all 360 scenes, rather than padding its budget with duplicates. The three conditions have 6,480 logical slots and share 2,278 distinct executions. Baseline has 29 complete candidate specifications and repair 35. Non-sphere ordered lists are unchanged; sphere additions alter the nearest-anchor neighborhood. Candidate identity contains runner, anchor, derived-control offsets, hold extension and native thresholds. Staged schedules have pre-position, finger ramp/hold, thumb ramp/hold, lift and support phases. Hold schedules retain each anchor’s settling/contact/grip stages and any micro, bridge, cradle or feedback phases, followed by lift,

support ramp and hold. Full numerical phase/control arrays are preserved in hash-locked anchor JSONs; no phase is refitted for V4. The two studies use the same parametric family generator with different frozen

Table S10: Realized geometry and mass ranges in the actual new V3 360 and V4 120 scene manifests, not substituted from an older 20-case suite. Box sizes are half-extents; curved objects use radius and axial half-length. All lengths below are mm and masses g.

Split	Family	N	Size 1	Size 2	Size 3	Mass
V3	bar	60	14.01–19.00	35.02–48.65	16.01–21.92	76.02–84.19
V3	capsule	60	13.71–18.29	23.19–37.91	0.00–0.00	60.49–66.73
V3	cube	60	22.01–28.23	22.01–28.23	22.01–28.23	55.01–64.35
V3	cylinder	60	18.07–23.44	27.01–36.98	0.00–0.00	65.08–69.88
V3	flat box	60	24.09–33.92	24.51–33.95	8.11–13.90	63.72–85.50
V3	sphere	60	17.04–23.95	0.00–0.00	0.00–0.00	51.93–56.77
V4	bar	20	14.37–18.80	35.54–48.68	16.01–21.83	76.32–84.21
V4	capsule	20	13.62–18.43	23.13–37.05	0.00–0.00	61.02–66.19
V4	cube	20	22.20–28.39	22.20–28.39	22.20–28.39	55.30–64.58
V4	cylinder	20	18.03–23.38	27.88–36.50	0.00–0.00	65.63–68.78
V4	flat box	20	24.03–33.95	24.20–33.07	8.19–13.81	65.25–84.11
V4	sphere	20	17.19–23.97	0.00–0.00	0.00–0.00	52.03–56.78

seeds. V3 seeds are 2026093031–2026093033; V4 uses 2026100202 and records input-selection randomness before scene generation. Sizes and XY positions are drawn uniformly within family-specific ranges, but mass is a deterministic function of sampled size and height is coupled to the geometry: the lowest point starts 20 mm above the floor. Thus these are not independent mass/size/height factors or unseen object categories. Bar XY ranges are $[-2.5, 8.5] \times [77, 84.5]$ mm; other families use $[-9.5, 9.5] \times [75, 98.5]$ mm. Yaw is uniform within ± 0.25 rad for cubes, ± 0.18 for bars and ± 0.30 for flat boxes; rotationally symmetric families have zero initial yaw. Contact friction parameters remain (0.9, 0.015, 0.004). For the free-object apparatus, mass is uniform 50–100 g; half-extents are 25–35, 22–29 and 25–35 mm; pad friction is 0.55–0.90. Nominal jaw command is 10–16 mm, with nine deliberately weak hand-only initializations at 3.02–3.15 mm and necessary-floor command fixed at 3.1 mm. Inactive-floor gap is 10–14 mm and the boundary group uses 0.7 times that gap. Every third initialization uses either a lateral offset within ± 7 mm or tilt within ± 0.15 rad and angular speed within ± 1 rad/s. All values and their coupling are listed in the frozen input design.

S5.1 Preserved V3 Comparisons

The following original tables and figures retain V3’s outcome definitions, model pair and candidate budgets. Environment-clear means that study’s literal no-contact gate; it is not retrospectively renamed no-external-load.

Table S11: Same 120 scenes and frozen candidates: legacy runtime geometry edits versus per-case compiled primitive physics. Counts are coverage at $K = 6$. Gains/losses compare physical conditions, not repaired versus baseline policies.

Program	Metric	Legacy	Compiled	Gains	Losses
baseline	Accumulated	37	38	7	6
baseline	Contiguous	37	37	6	6
baseline	Clear contiguous	18	13	0	5
repair	Accumulated	43	43	7	7
repair	Contiguous	43	42	6	7
repair	Clear contiguous	24	18	0	6
no-extra	Accumulated	37	38	7	6
no-extra	Contiguous	37	37	6	6
no-extra	Clear contiguous	18	13	0	5

Table S12: Three new, fixed-size test blocks under compiled primitive physics. Each family contains 60 scenes; all rows use $K = 6$. B: baseline; R: repair. Acc., cont., clear: original accumulated, contiguous, and environment-clear contiguous gates.

Family	Acc. B	Acc. R	Cont. B	Cont. R	Clear B	Clear R
Cube	49	49	49	49	0	0
Cylinder	35	35	34	34	34	34
Bar	21	21	21	21	0	0
Sphere	2	15	2	13	2	13
Capsule	1	1	1	1	1	1
Flat box	20	20	15	15	0	0
Total / 360	128	141	122	133	37	48

Table S13: Fixed candidate prefixes on all 360 new scenes. Selected-clear scores the unchanged original outcome-aware selector; clear coverage is the existence of a passing candidate under the corrected contract.

Program	K	Accumulated	Contiguous	Clear coverage	Selected-clear
baseline	1	48	46	11	11
baseline	3	111	106	27	27
baseline	6	128	122	37	37
repair	1	52	49	14	14
repair	3	122	115	36	36
repair	6	141	133	48	48

Table S14: Replication and anchor-removal ablation under the clear-contiguous contract ($K = 6$). Each new block was fixed at 120 scenes before execution. Gains/losses are paired program differences within each block.

Block	N	Baseline	Repair	No extra	Gains	Losses
Block 1	120	11	15	11	4	0
Block 2	120	13	17	13	4	0
Block 3	120	13	16	13	3	0
Combined	360	37	48	37	11	0

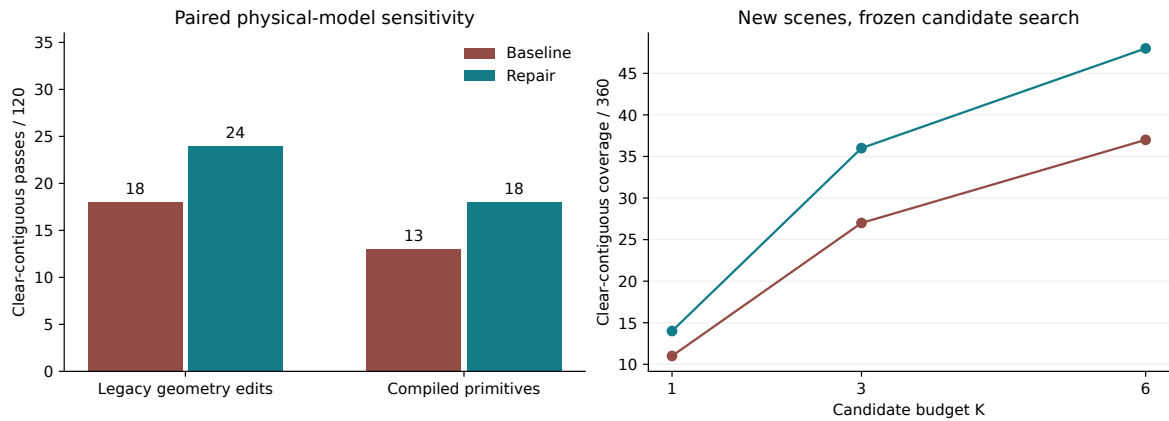


Figure S2: Preserved V3 model and budget comparison. Model changes were not an isolated inertia intervention. Historical intervals and denominators differ from V4 and are not pooled with it.

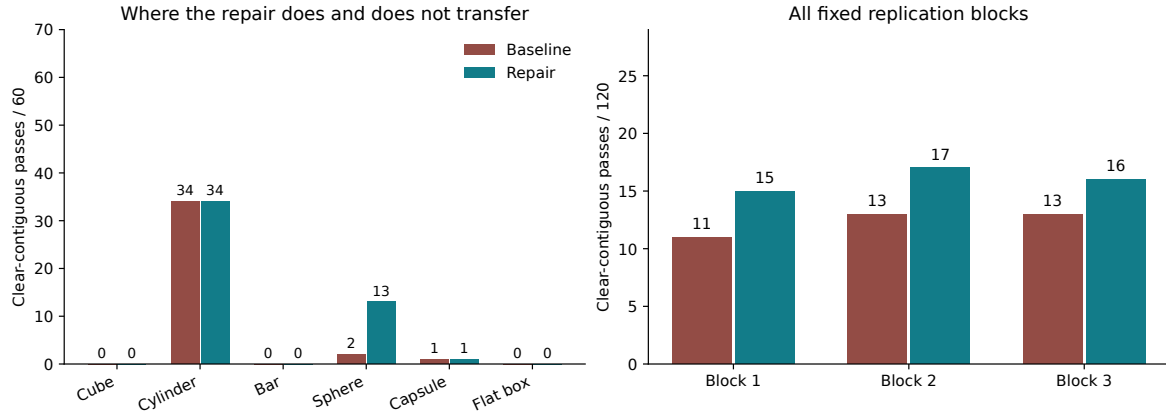


Figure S3: Preserved V3 family-level results. The 11 no-contact gains occur in spheres; this does not establish broad family generalization.

S6 Numerical Qualification and Recovery

The observer reads each physical contact solve without inserting a forward solve into the live trajectory. Target-directed wrenches use the geom-2 convention of `mj_contactForce`. Contact geometry/forces belong to the pre-integration solve state; post-integration poses and velocities have separate fields. Integer step indices and measured timestamps are retained without decimation for adjudication. Full snapshots include warm starts, controls, applied forces and mocap; both branches consume the same input tape. The branch gate uses prefix conditions and local Q , not future native success.

The 80 calibration controls contain 25 no-load reference passes and 55 failures; B4/B5 match all 80. Development branches have 57 both-retain, 10 dependent and 13 unreachable states. Thirty-two executions of the first eight controls vary timestep and solver iterations; they are repeated numerical conditions, not 32 new scenes. A selected natural 20-scene development subset has 17 dependent and three both-retain late states, with pre-branch load in the latter. All 20 labels remain unchanged under the registered numerical settings. These selected exclusions do not estimate natural prevalence. A simple pre-branch load threshold separates that subset, so no superior learned dependency predictor is introduced. Four additional single-burst development cases show 16.9s accumulated versus 9.9s continuous unloaded Q ; they are not added to the 80-case calibration denominator.

The five Allegro geometry failures were retained using a post-failure, fail-closed bookkeeping recovery. Its hash is separate from the input-only protocol lock. No threshold, physical parameter, candidate, data role or primary comparison was changed. Invalid telemetry and physical indeterminacy are separate fields that can overlap; neither may be silently recoded as a verified failure. Natural references were simple random samples (30/120 rank-1 Allegro entries; 30/100 for Fetch and DGB), not selected successes.

S7 Offline Reproduction and Raw-Evidence Scope

The label evidence archive contains execution scores, reference labels, all 720 candidate future endpoints, the pre-future choice seal, protocol and source manifests, operating-point labels, code, recovery provenance and claim ledger. Its protocol SHA256 remains `ffc60fe94dca45e826c2f864865cbeb7c749b7a11b6f05784b9c6f24b896b9a0`. The frozen scientific files are unchanged; V4.5 adds only manuscript, figure-composition and packaging files.

Extract the label evidence archive. From its top-level directory, run:

```
python -B reproduce_v4/reproduce.py --out ../v4-recomputed
```

Use a new output directory. Python 3.11, NumPy and Matplotlib are required. The original verifier blocks MuJoCo imports and network connections, verifies packaged hashes before and after computation, recomputes source tables and paired intervals from individual labels, reconstructs U2 from sealed choices and future endpoints, and checks operating curves and the claim ledger. It also reconstructs the three original V4 figures and rescores three included primitive-control raw examples. It does not replay physics.

The V4.5 publication-only figure command is separate:

```
python -B scripts/v45_editorial.py figures
```

It uses the same recorded JSON/CSV and saved display arrays, checks the displayed counts, and composes the three main-paper figures. The four-cell example table comes from recorded control outcomes, not a new simulation. The selection stacks classify failures only after checking the corresponding future status is valid; unclassified returned outcomes cause an error.

What is supplied. The label package includes all individual scored records needed for the reported aggregates. The separate core-raw ZIP includes all 240 nominal final primitive-control NPZs and their eight margin/gap partners, with a dedicated hash/size manifest. These controls contain original primitive apparatus models, not redistributed third-party hand meshes. The companion is optional for label-level reproduction; it is additional source evidence, not a claim that all 248 raw records have been independently re-adjudicated during this editorial release. The original verifier’s three raw rescoring examples remain the verified raw-reconstruction scope.

What is not supplied or claimed. Most natural-task, calibration and retrospective raw NPZs remain in the local archive. The full raw inventory records hashes and sizes, but an inventory is not raw access. No public host for the complete archive is asserted. Thus aggregate reproduction, access to controlled raw evidence, full-matrix raw adjudication, and dynamic reproduction are distinct levels. No complete independent raw audit or clean-machine simulation rerun is claimed.

Main experiments used CPython 3.11.1, NumPy 2.4.6, MuJoCo 3.8.0 and SciPy 1.17.1; DGB used NumPy 1.26.4 and MuJoCo 3.3.2. Offline release checks used NumPy 1.26.4 and Matplotlib 3.11.2. The preserved first-case dynamic checks cover all four sources in existing isolated environments; they are not fresh installations or complete-matrix reruns. V4.5 adds no dynamic check.

External provenance. FetchExpert [1] is pinned to commit `3e2dd8c2e5f24702081867d82a78eb1264534f94`, file `experts/fetch_expert.py`; Gymnasium-Robotics [2] is version 1.4.2 (FetchPickAndPlace-v4). DGB is pinned to `d9ea6cf282de1f463c20fa54b4f68d7025bad40e`. Dependency records list component-specific licenses and hashes. Third-party model assets and the unlicensed DGB checkout are excluded; missing assets are not silently downloaded, substituted or reported as dynamically verified. No new license grant for the author’s source is made by this local package.

S8 Disclosure

OpenAI Codex assisted implementation, experimental workflow, analysis tooling, figure composition and manuscript preparation under author direction. The author is responsible for all claims and the final submission. No controlled comparison estimates the assistance’s causal effect.

References

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- [4] Google DeepMind. MuJoCo Menagerie. https://github.com/google-deepmind/mujoco_menagerie, 2026. Pinned robot assets and hashes are listed in the artifact dependency manifest.
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